

Sublevel Caving

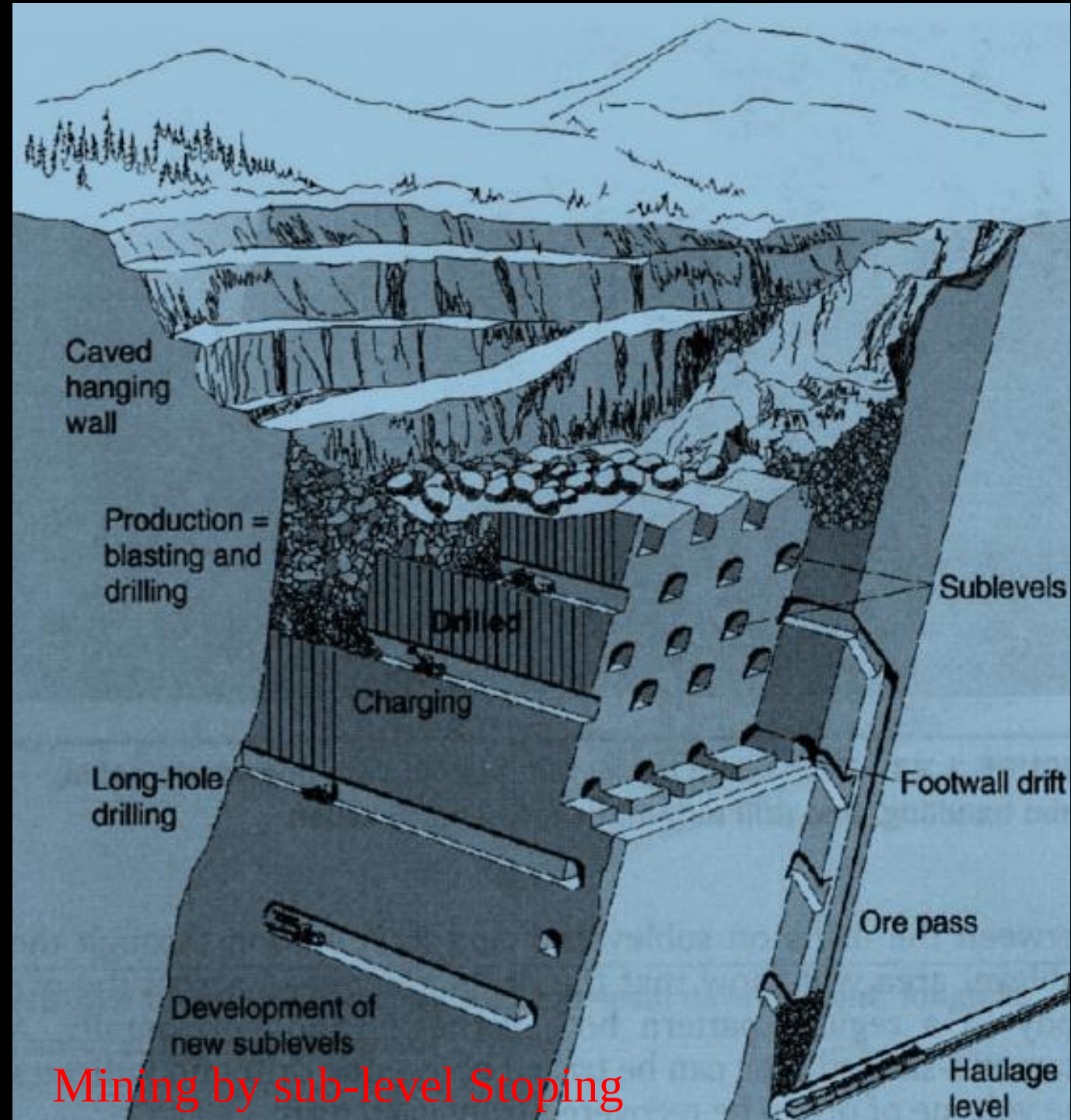
13/01/2020

Applicability I

- *Sublevel caving is a mass mining method based on utilization of gravity flow of blasted ore & caved waste rock.
- *Sublevel is a high production, bulk mining method.
- *Sublevel caving is a caving method, causing subsidence at surface.
- *Sublevel caving is typically deployed under ground method, when surface mining of the deposit is no longer economical & subsidence at surface is allowed.

Applicability II

- *In opencast mining system, lease is acquired for ore body along with surface right.
- *This makes it enable to go for underground sub level caving, where subsidence can be permitted.
- *Sublevel caving is the cheapest underground production method with drilling & blasting.



Ore Body Dip I

- *Sublevel caving method is applicable to large, steeply dipping regular ore bodies.
- *Ore body should have dip more than 40 degree.
- *With reducing dip, recovery of ore goes down & dilution of ore goes up resulting higher loss of ore.
- *Foot wall contact should be smooth & exceeds angle of repose of broken ore & rock.
- *This shall permit gravity flow of blasted ore through draw point.
- *Very thick horizontal or near horizontal ore body can be mined by sublevel caving.

Ore Body Strength I

- *Ore strength can be weak, medium hard or strong,
- *HW of ore body should be weak to moderately strong & cave able.
- *FW can be strong, self supporting, competent to moderately strong.
- *Preferable ore strength is 55 Mpa, but stronger or weaker ore body can be mined by this method.
- *Variation of rock strength has to be compensated at design state.

Ore Body Shape I

- *Deep-seated ore body can be mined by Sublevel caving.
- *In ore body less than 6 m width, sublevel stoping is not performed because of higher cost and larger dilution.
- *In wide ore bodies, transverse stoping is applied other wise longitudinal method of stoping is followed.
- *Ore body with even. regular & fairly well defined boundary cause less dilution.

Ore Body Shape II

- *Waste inclusion in ore body will caused dilution of ore as because mining method is not selective.
- *Irregular ore body causes high level of dilution.

Development I

- *Mine development starts from a shaft sunk in foot wall to avoid any subsequent caving effect from stopes.
- *Cross cuts are developed from shaft to reach ore body.
- *Main ore pass, feeding ore to shaft hoisting system is connected to these cross cuts.
- *Cross cuts are main entry to stope, size of cross cut is decided based on size of equipment to be taken to stope.
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Development II

- *Vertical height of stopes is same as that of cross cut height interval.
- *All stope ore pass leads to main drive for further hauling of ore to main ore pass.
- *Generally haulage levels are at 60 to 80m vertical interval.
- *Stope height is also 60 to 80m.
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Longitudinal Stope Development I

- *Access raises driven in ore body are used to further subdivide ore body into blocks for stoping.
- *Ore body is divided by raises at 60 to 90m length along strike direction.
- *Raises are excavated to connect two level, known as service raise or main raise.
- *In case where track less machinery are used, ramp is excavated in between two levels.
- *Sublevel drifts are developed in all levels at uniform pattern & equal vertical interval as far as possible.
- *Sublevel vertical interval varies from 10 to 30 m.

Longitudinal Stope Development II

- *Sublevel mine drifts are primary working places for stoping.
- *Width of drift varies from 5 to 6m & height of drift 3.5 to 4m.
- *Size of drift is practically decided by choice of machine & ore flow characteristic.
- *More than one sublevel may be used at each levels, depending on width & geometry of ore body.
- *Wider ore body shall have parallel sublevel drift to cover entire width of ore body.

Longitudinal Stope Development III

- *Each sublevel drift is connected to transportation drift having suitable width & height to accommodate mine truck.
- *In competent rock, sub level drift does not require any support.
- *In less competent rock mass, stability is achieved by a combination of smooth blasting, shot creting & rock bolting.
- *Use of rock bolt at sublevel drift should be minimum, rock bolts creates nuisance at time of mucking.
- *Transport drift is usually located at foot wall waste rock at each sub level.

Longitudinal Stope Development IV

- *Sublevel drifts are connected to transport drift in each level through ore pass, raise or ramp.
- *Ore pass is developed to transport ore & waste rock from transport drift to main haulage level.
- *Main haulage is located in footwall waste rock

Longitudinal Stope I

- *An expansion slots is cut by stripping raise from footwall to hang wall of ore body with help of parallel holes.
- *Expansion slots provides free space so that first round of blast gets required free face.
- *Over all stoping is carried out over hand to facilitate ore flows in to lower sub level.
- *Stoping starts by blasting upper most sublevel, then moves progressively down wards.
- *Lower levels are in development mode & normally all development is concentrated at one level for better machine utilization.

Longitudinal Stope II

- *Blast face is kept slightly inclined to outer side.
- *Void created by removal of broken ore is filled up by caved hang wall waste & cap rock.
- *Hang wall has to cave after removal of ore, in some cases caving may have to be induced by blasting.
- *Upper level blasting & mucking remains ahead of lower level.
- *Drilling operation is in progress at sublevel below.

Longitudinal Stope III

- **While mucking is in progress at top most sublevel horizon of stope, fan holes are blasted at intermittent levels below.
- *Extraction limit is maximum acceptable amount of dilution.
- *In sub level caving all ore must be drilled & blasted in order to utilized gravity flow of fragmented ore.
- *Blasted muck is mucked (loaded) at lower level by LHD & LHD dumps ore to ore pass leading to main haulage level.

Longitudinal Stope IV

- *Extraction of ore is carried out through back (roof) of drift, back of drift works as draw point.
- *Blasting & drawing ore starts simultaneously in staggered fashion.
- *Most modern system uses LHD (tire Mounted) loading directly from ore drawing location at mining sublevel.
- *Remote controlled LHD has proved most successful & safe for loading at sublevel under exposed condition.

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Longitudinal Stope VI

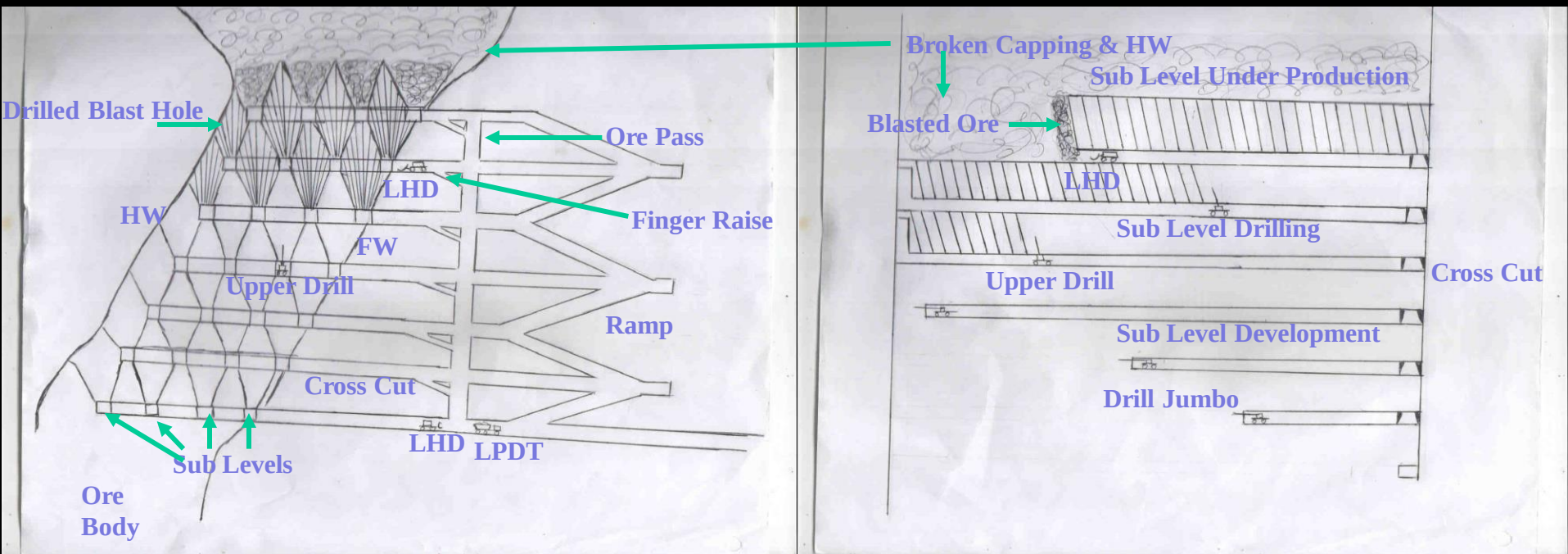
- *Distance between main haulage drift & stope is kept in such a way, so that main haulage does not get effected by stoping in future.
- *At haulage level ore is transported by LPDT / LPET to main shaft ore pass for hoisting to surface.
- *Track mounted rail transport is used, where ore has to be transported long distance before dumping shaft ore pass for hoisting to surface.

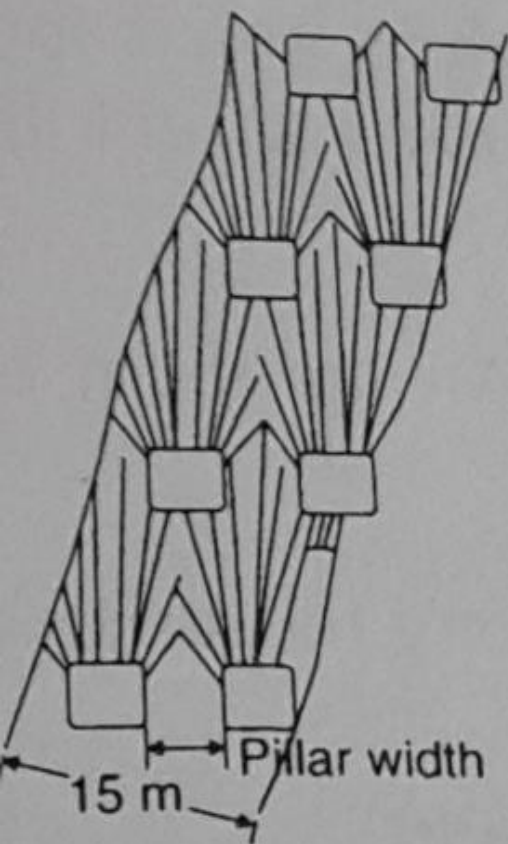
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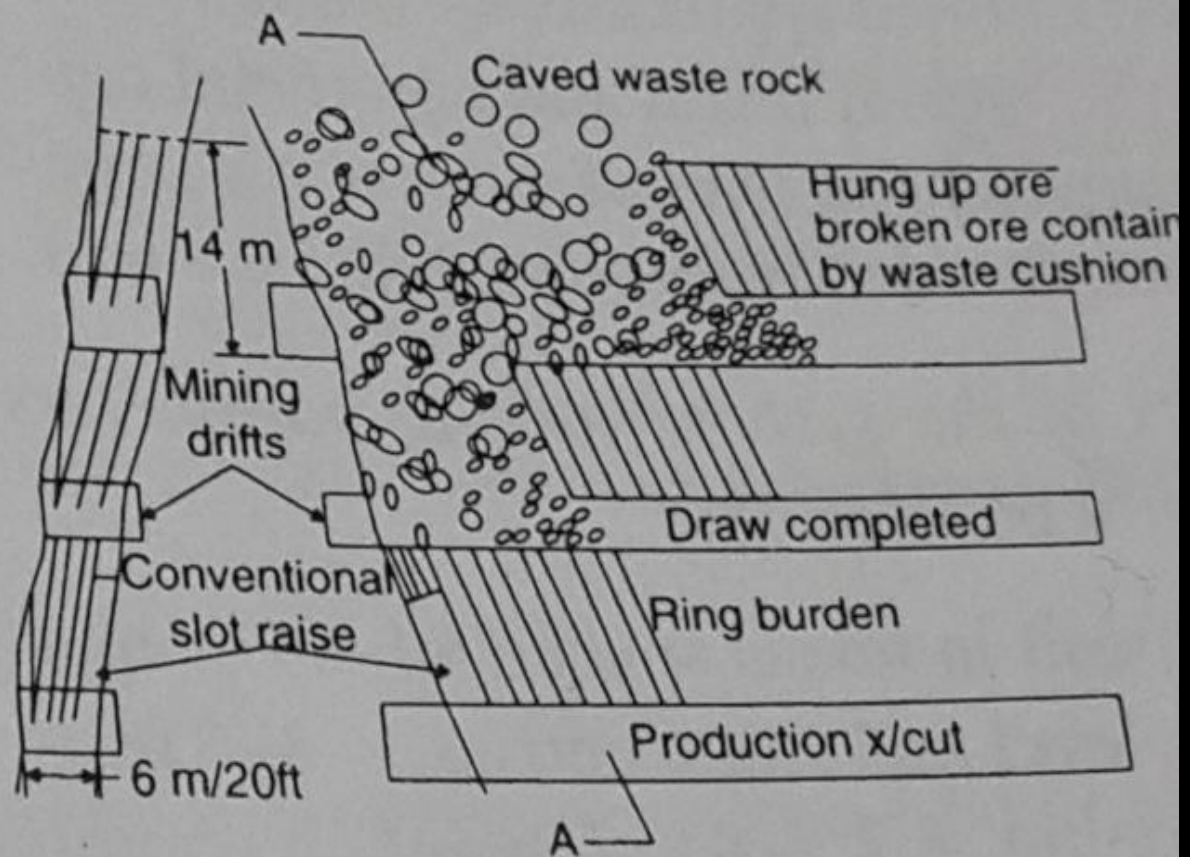
Mechanised Sublevel Caving *

with LHD, Drill Jumbo



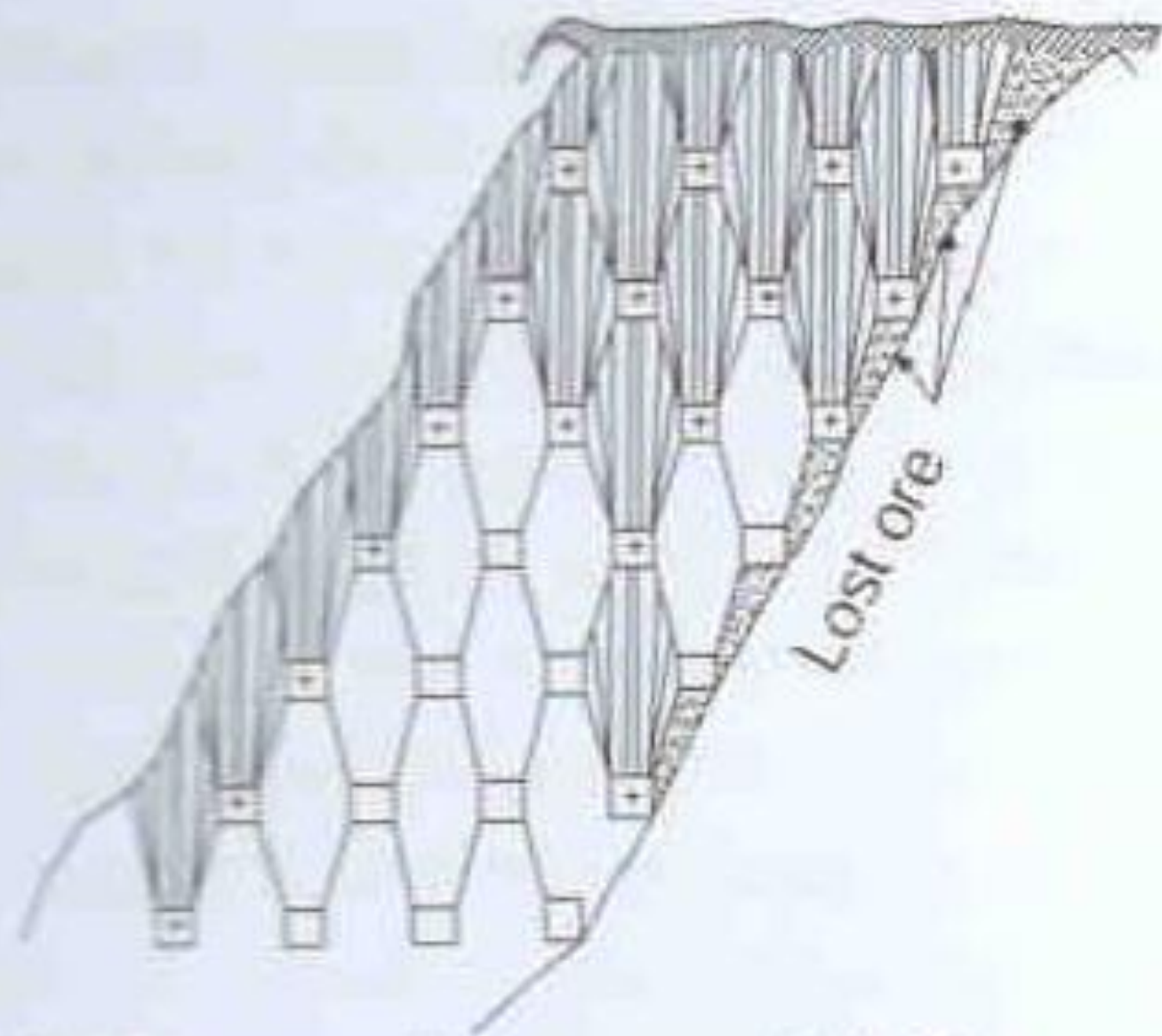


Cross-section along A-A

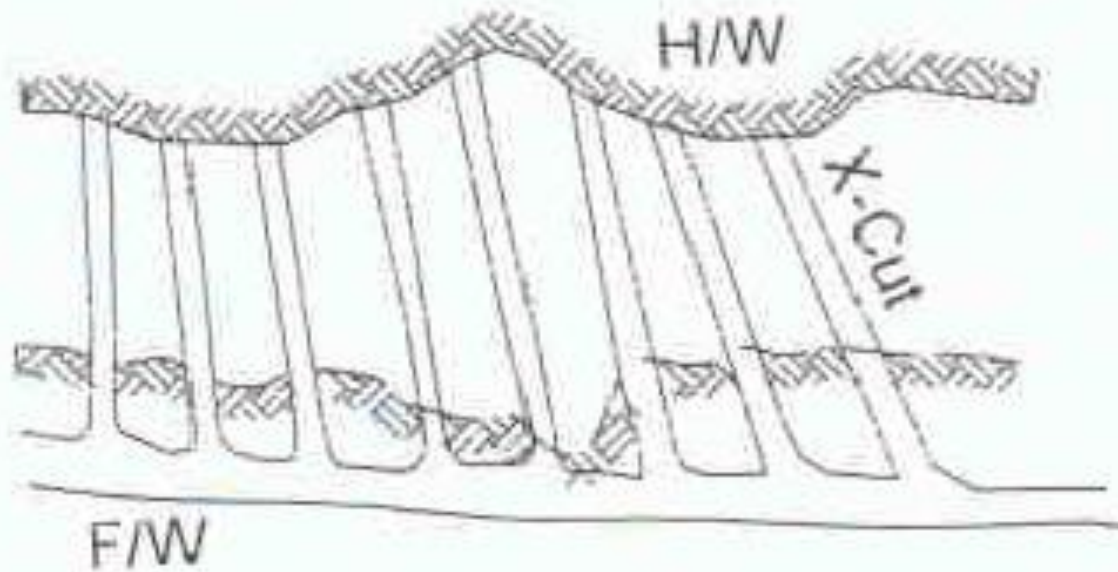


Vertical longitudinal projection

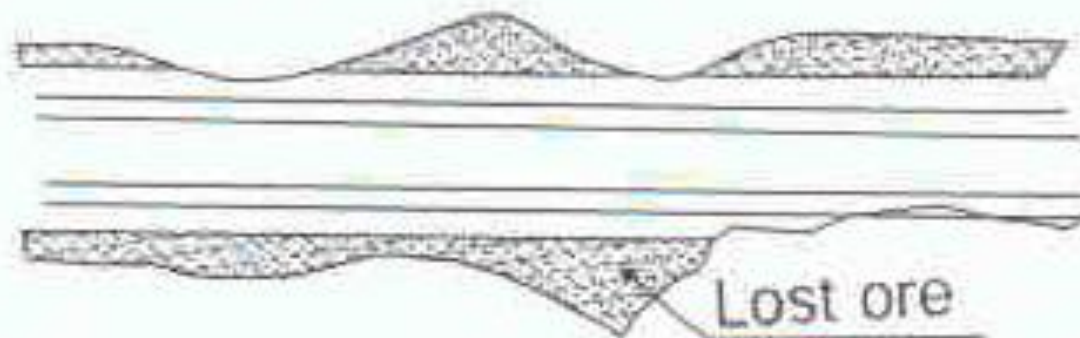
b) A typical longitudinal sublevel-caving stope



(a) With steep dips recovery is better

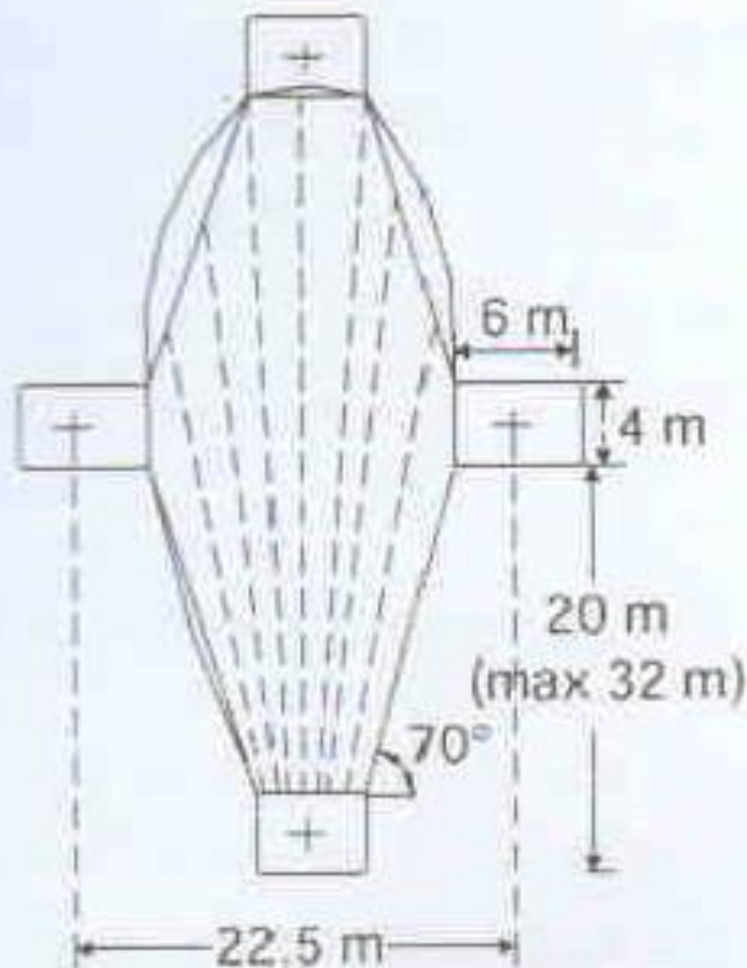


(b) In wider orebodies (Tr. S/L Caving)
ore losses are little



(c) In narrow orebodies (Long. S/L Caving)
ore losses are more

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(d) Relative positions of sublevel horizons-for large dimensioned stopes with the use of 110 mm. dia. blastholes at LKAB, Sweden

Drilling & Blasting 1

- *Size distribution & shape of fragments largely depends on density of drill holes & on blasting parameters.
- *Purpose of drilling, blasting is to obtain coarse material with fragmentation that will permit gravity flow & undisturbed ore extraction.
- *Aim of drilling & blasting is to achieve the purpose with least effort.
- *Ore should be broken just to right size at minimal drilling, blasting cost.

Drilling & Blasting I1

- *Course material is characterized by different size distribution & shape.
- *Some time, coarse material can be very heterogeneous.

Drilling I

- *It is required to established drill ability of ore, required fragmentation, size, hole size, hole length, orientation, spacing of drill holes and drilling accuracy for drill hole pattern design.
- *Drill hole pattern is decided based on these information.
- *Production form sub level stope is by blasting long holes.
- *Machines are mounted on column and arm.
- *Production holes are drilled in up ward direction from sublevel.

Drilling II

- *Production drilling holes are of 50mm to 70mm diameter where sub level vertical height distance is 10 to 15 m.
- *For sublevel with vertical distance 15 to 30m, hole diameter is 100mm to 110mm.
- *Hole spacing is 1 to 1.5m
- *Hole burden is 1 to 1.8m
- *Mine development is done by drill jumbo, jack hammer with leg, stopper.
- *Production drilling (fan or parallel) is done by single or double boom drilling rig suitable for drilling upper holes.

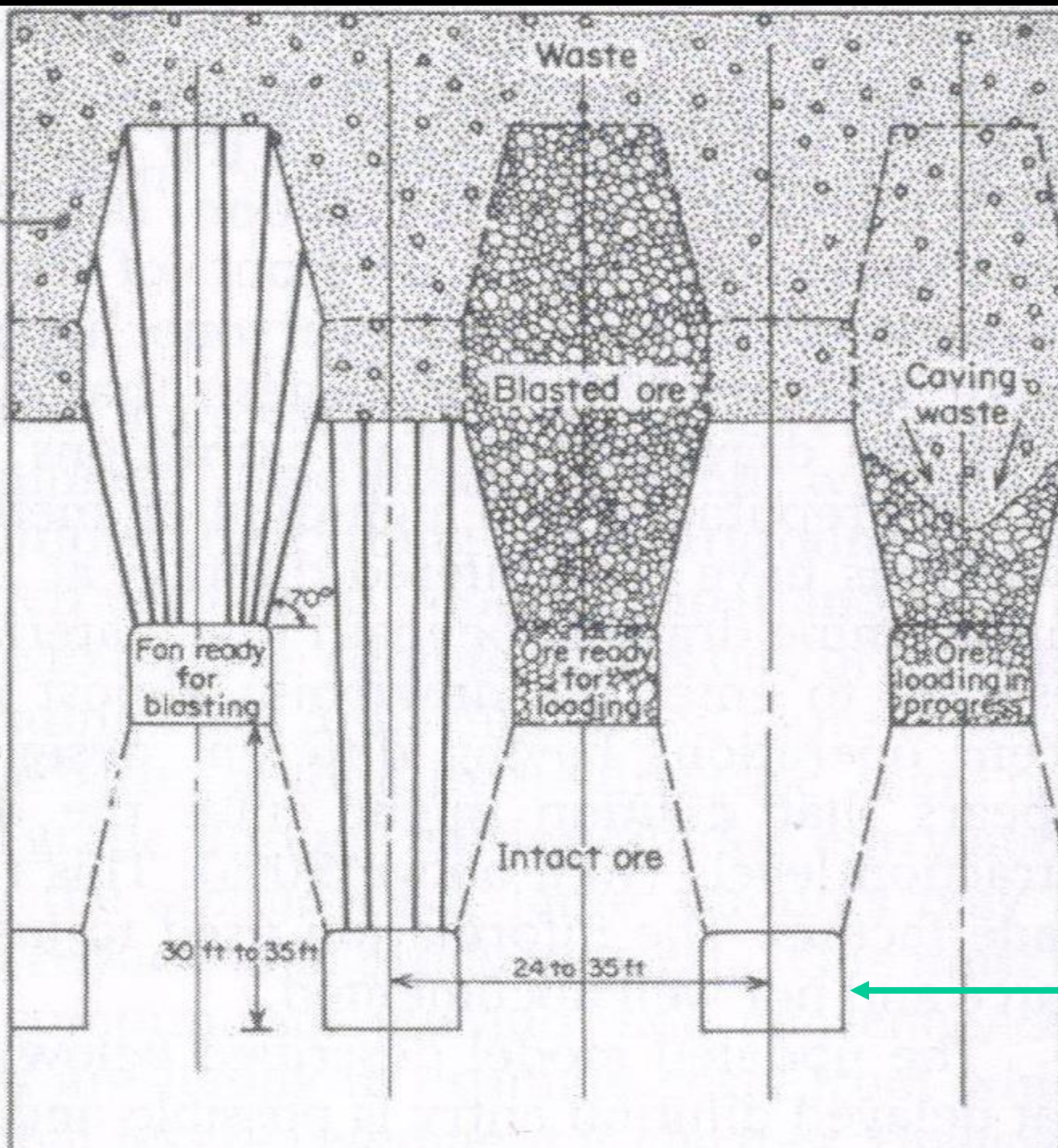
Drilling III

- *In Sublevel caving method fan cut pattern is followed, holes are drilled vertically up wards from sublevels.
- *Fan pattern has inclination of 70 to 80 degree out ward (to wards slot raise).
- *Inclination of outer hole in fan cut is 70 to 85 degree.
- *Rotary percussive drilling system can drill up to 110mm dia hole up to 50m length with less than 2% deviation.
- *Drilling accuracy has economic implication in mining operation.
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Drilling IV

- *Fan cut or parallel hole are drilled for sub levels which are up to maximum 30m apart.
- *Parallel holes are drilled for steeply dipping & relatively narrow deposit.
- *Fan pattern drill is followed in wider ore bodies.
- *Drilling pattern and hole sizes are frequently established by pre setting power factor to size of ore block to be mined.
- *Typical hole spacing varies from 1m to 1.5m and burden varies 1m to 1.8m.

Production Drilling



Sublevel & Draw Point for Broken ore.

Blasting I

- *Selection of explosive is done based on based on requirement in fragmentation, hole size, spacing and burden, holes conditions, water inflow, allowable blast size, hardness etc.
- *ANFO is lest expansive explosive.
- *ANFO is to be pneumatically loaded in fan holes.
- *ANFO can not be used, where water flow is problem.

Blasting II

- *ANFO can not be used in watery hole for a period more than few hours.
- *Emulsion explosive is be used for watery hole.
- *Emulsion explosive is water proof, high density and typically high velocity explosive.
- *Emulasion explosive can be loaded in down holes or up holes by equipment (SME).

Production Blasting III

- *Decision to load a hole with either a column charge or deck charge is based on consideration such as allowable powder factors, hardness of ore, up holes and down holes conditions, fragmentation requirement, blast vibration.
- *Blasting is sequential, so that one ring of holes is fired at a time,
- *Rings are blasted in succession in to opening created by blasting of previous ring.

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Production Blasting IV

- *Timing sequence of blast is critical to success of blast and resultant fragmentation.
- *Proper blast timing is critical in minimizing the effect of excessive blast vibration and concussion effect underground.
- *Secondary blasting is necessary if primary blasting has failed to sufficiently fragment ore to that can be dealt with ore removal and hoisting equipment.
- *Plaster shooting (Shaped charge) involves placing explosive on top of bolder and boulder is shattered by detonating explosive.

Production Blasting IV

- *Secondary blasting is an expansive, time consuming exercise.
- *Close engineering control on primary drilling and blasting will minimize frequency of secondary blasting.
- *Blasting cost is 10% of total mining cost and drilling costs 30% of total mining cost.

Transverse Stope I

- *In case of transverse stoping, stopes are divided transversely at 50 to 60m length.
- *Distance between two sublevels vertically varies from 10 to 30m.
- *In transverse stoping, space in between cross cut is 8 to 23m.
- *Development work first starts at top sublevel.
- *Sublevel development includes drive development through out length of stope in waste rock at location at least 10 m away from foot wall contact of ore body.
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Transverse Stope II

- *From this drive, drilling cross cut, spaced at pre determined interval are driven across ore body up to its hang wall contact.
- *At hang wall contact, each of these cross cuts are connected resulting a drive (hang wall drive).
- *At hang wall, slot raise is driven to ultimately convert in to slot raise by widening with longh parallel holes.
- *These slots are used as free face for fan drill at drill cross cut
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Transverse Stope III

- * Similar development work is carried out at each sublevel horizon.
- * Ore pass driven between two level & thus connected to each sublevel's horizon to facilitate dumping ore up to main haulage level.

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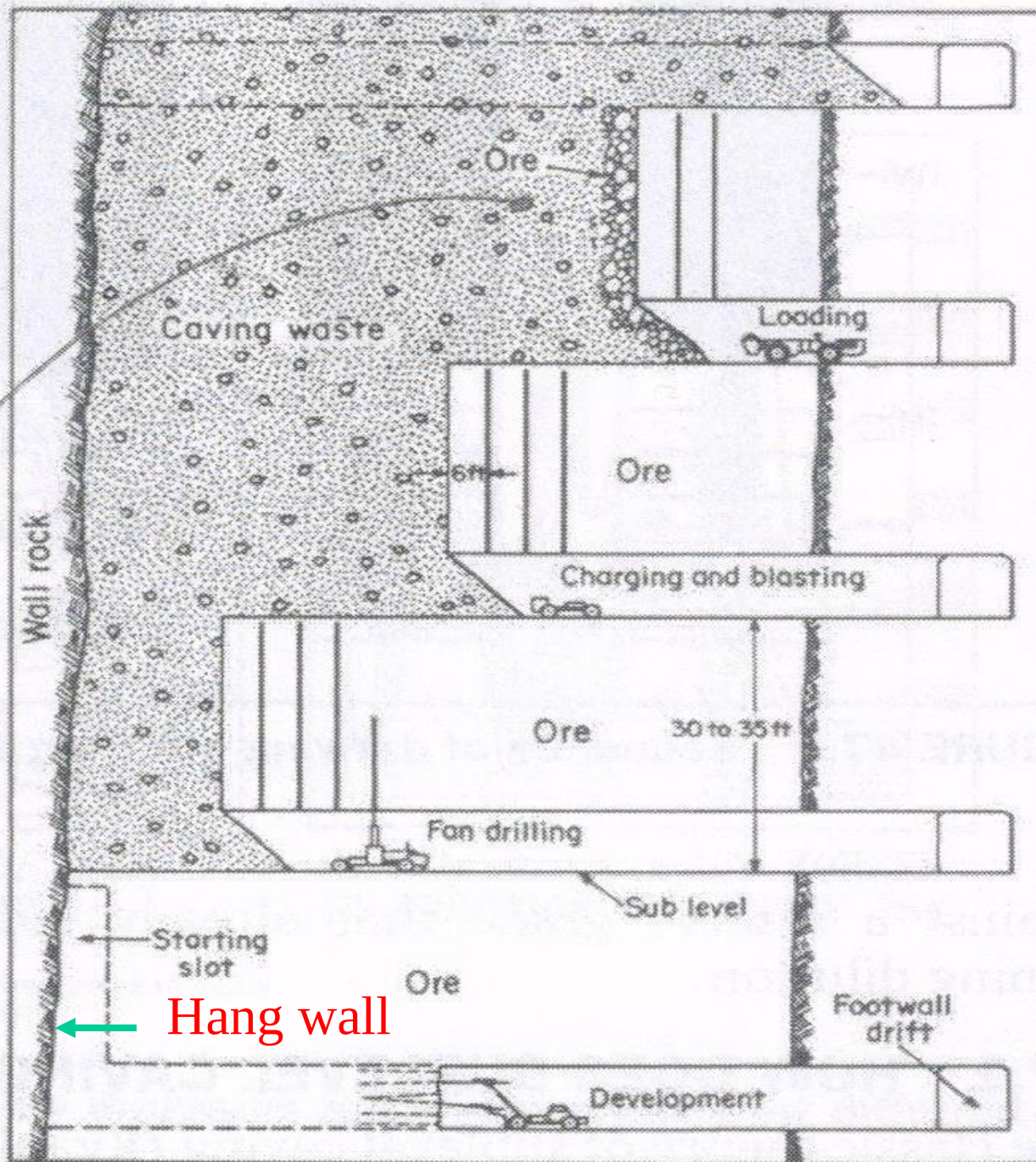
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Transverse Stope



Support I

- *Different type of rock bolts and cable bolts are used for local support.
- *Some time regional stress have been shown to effect stope design and dimensions.
- *Regional stress are to be managed by local support.
- *Resign anchor bolts are most successful, but cement grout bolts are also used.

Classification of Course Material I

- *Type I course material with large spherical pieces of more or less same size & shape.
- *Type II represents a material almost of same size but different shape.
- *Type III indicates a course material composes of large fragments, chippings & sand.
- *Type IV represents a course material that characterized as a mixture of large blocks, medium size fragments, chippings, sands, rock powder, earth, clay compounds etc.

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Classification of Course Material II

- *Type III & Type IV, course material could change considerably the mechanical properties & behavior, depending on percentage of fine particle (sand, clay etc) & of pressure of water.
- *If water is present, especially, in course material type IV, fine verticals could create plastic & sticky compound.
- *This could change very unfavorably its properties & behavior.
- *Below 40 degree inclination, course material requires mechanical transportation.

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Classification of Course Material III

- *With larger angle of inclination gravity transportation could be used in open chute or in ore pass.
- *Type I & II at 40 to 70 degree inclination, open chute can be used for transportation.
- *Type III & IV at 70 to 90 degree inclination, must be transported through steep ore pass.
- *Almost all blasted ores & water are similar to type III & IV.
- *Composition, properties, behavior etc of these material could be very different because they could consists of very large boulders down to finest particle.

Factors of Gravity Flow I

- *In short term, inclination of bottom (draw point) has no influence on gravity flow of its fill material.
- *Bottom must be steeper in order to influence flow in long terms.
- *Extraction opening at bottom should be of minimum dimension, but it should be sufficiently large for a free & un interrupted flow of material.
- *Zone which shows gravity motion is called active zone.
- *Material which is not affected by gravity motion is called passive zone.
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Factors of Gravity Flow II

- *Boundary of gravity motion is similar to an ellipse.
- *Boundary has a vertical & axial section of axis-symmetric body (elongated ellipsoid)
- *After extraction of certain volume, remaining material replaces this loss by loosening.
- *Size of ellipsoid (its eccentricity) increases as material is extracted.
- *Depending up on material properties & acting conditions, volume of ellipsoid of loosening can be about 14 to 16 times larger than volume of extracted material.

Factors of Gravity Flow III

- *Extraction of certain volume of material results in development of a certain ellipsoid of loosening.
- *At beginning, there is no motion at boundary of ellipsoid of loosening.
- *Highest practical velocity is at center of discharge opening.
- *After discharge of certain amount of material, there is velocity at boundary of ellipsoid.
- *Velocity increases from center to outside of ellipsoid envelop.
- *There is “0” particle velocity at perpendicular to vertical axis of ellipsoid envelop.

Factors of Gravity Flow IV

- *Particle velocity in ellipsoid envelop is same for a particular surface.
- *Boundary of ellipsoid of loosening has momentarily zero velocity.
- *Zone of similar elongated ellipsoid is called ellipsoid extraction.

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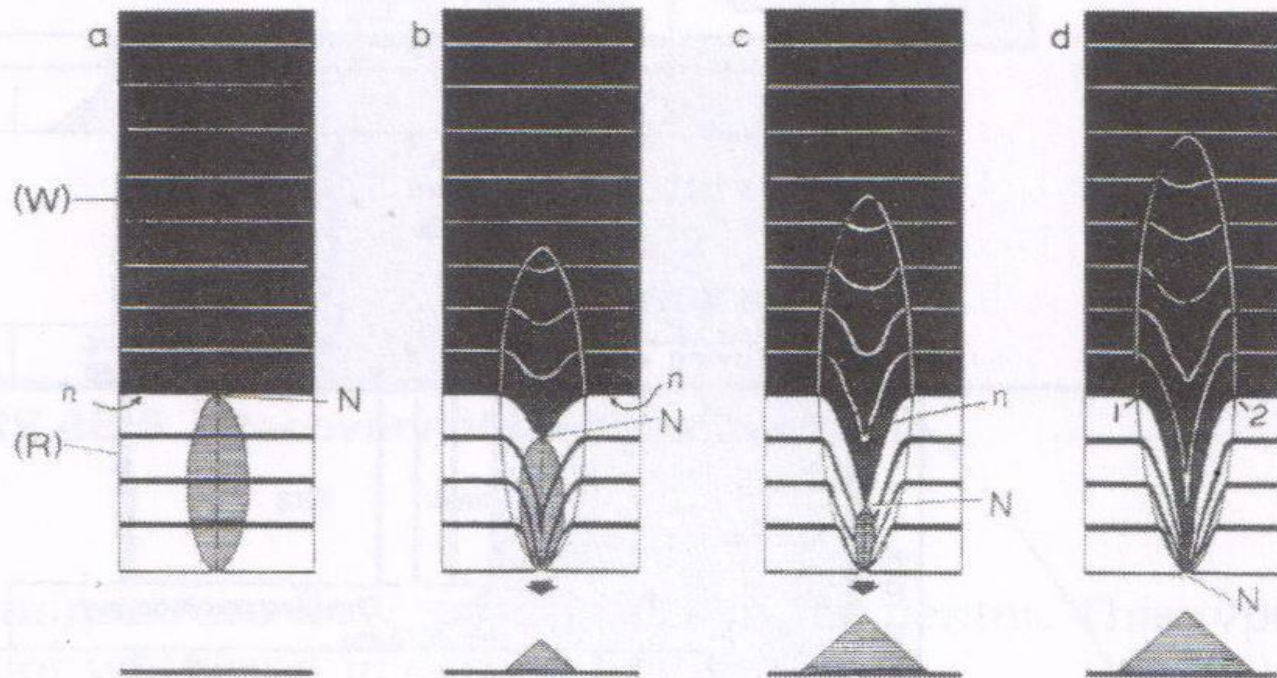
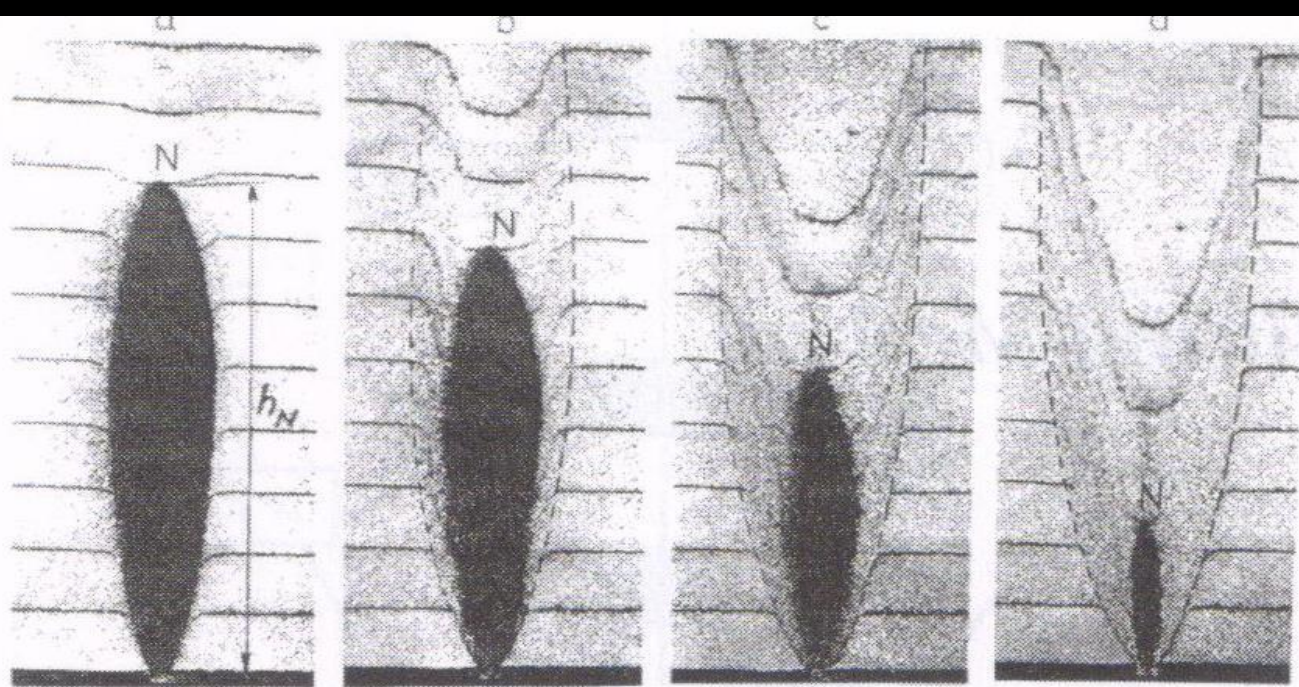
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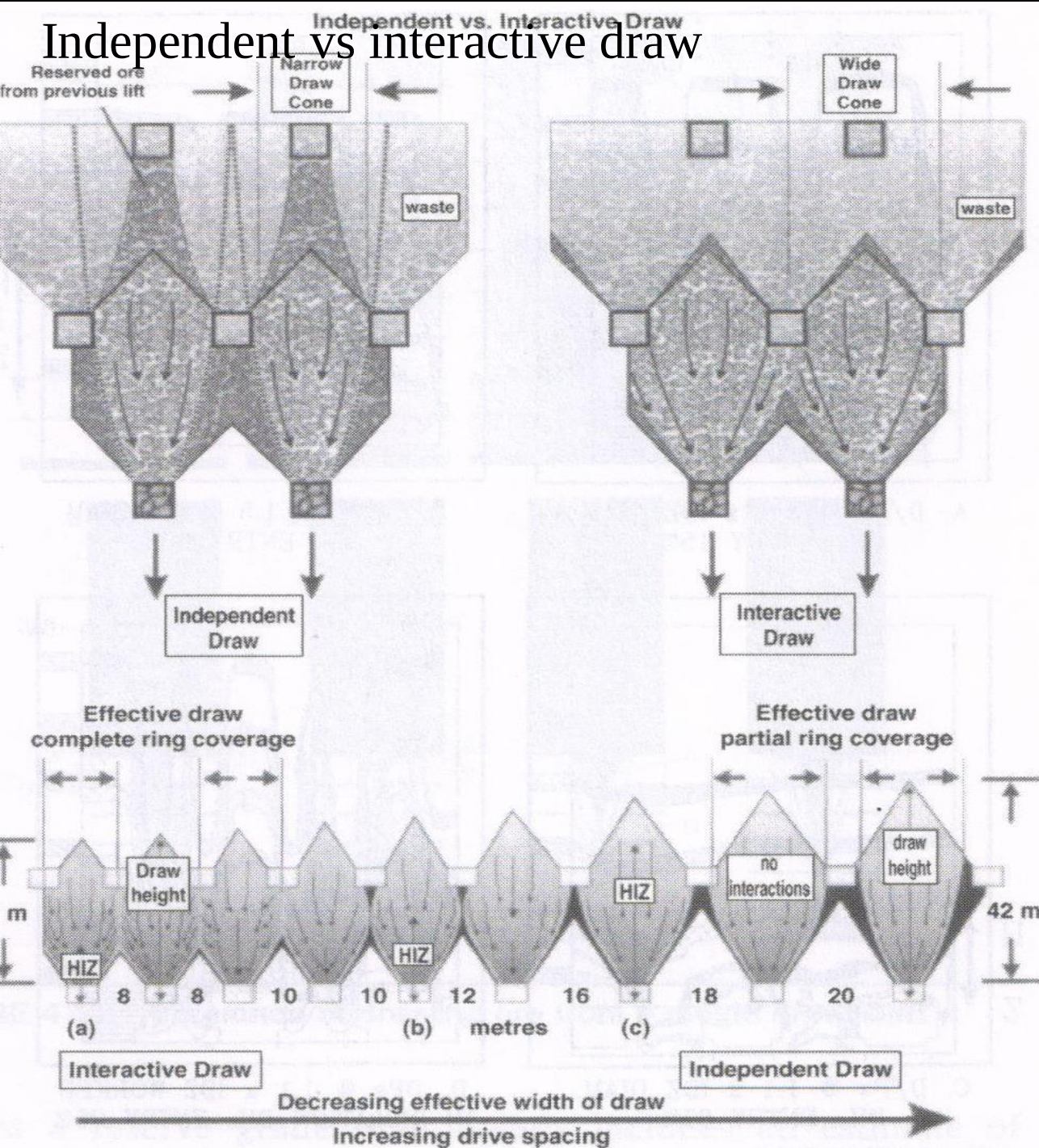
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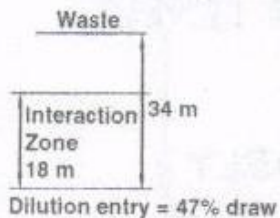
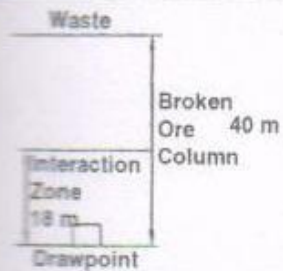
Ellipsoids of motion and isolated draw from a single draw point



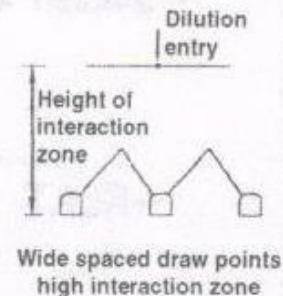
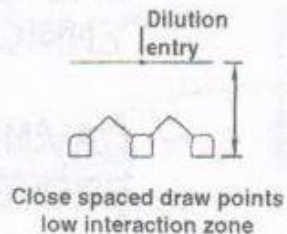
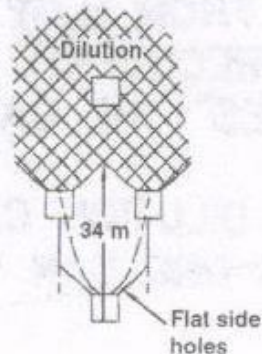
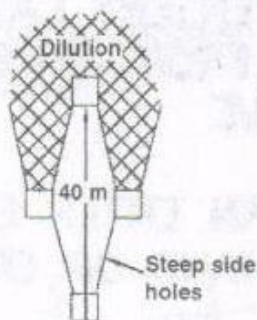
Independent vs interactive draw



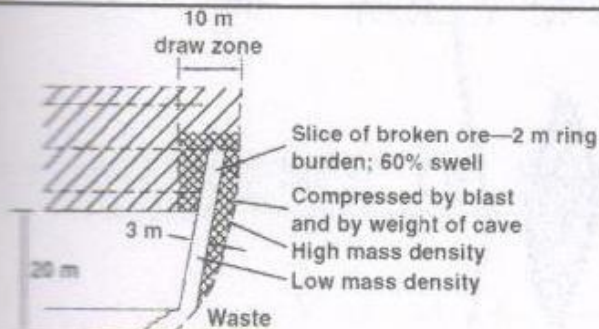
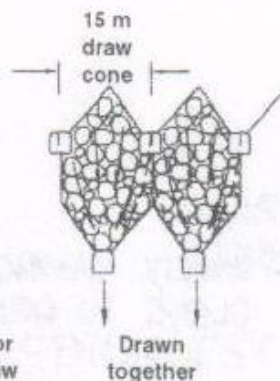
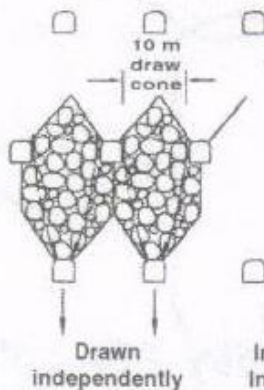
Interactive draw: the simultaneous drawing of adjacent draw points will result in a much wider zone of moving material. Thus one should try and draw from several draw points in a line.



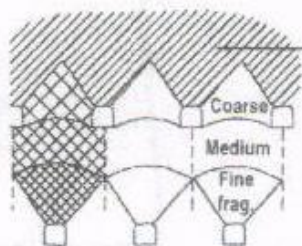
$$\text{Dilution entry approx.} = \frac{40-18}{40} = 100 = 80\% \text{ draw}$$



Difference in dilution entry with different heights of the broken ore column and interaction zone



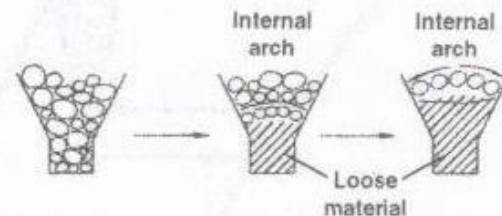
Draw cone diameter of 10 m



Configuration of Blasted Ore and Compacted Waste

High density, compacted waste

—Fragmentation varies through the ring profile—coarser at the top and fine near the draw point. Fine material gets drawn preferentially. The coarse holds back the waste.



Formation and breakage of arches within the broken mass : the coarser the fragmentation the wider the arches and the wider the draw

Advantages I

- *Sublevel caving is very amenable to mechanization, stoping efficiency is very high.
- *Sublevel caving method can achieve high rate of production.
- *Productivity of sublevel stoping is very high.
- *Sublevel caving is safe because it is operated from small opening, hence low human exposure.
- *Sublevel caving is easy to ventilate.
- *Sublevel caving is highly flexible to suite market condition & geometry of ore body.
- *Ore production starts immediately after blasting.

Advantages II

- *In Sublevel caving job is concentrated at limited places, providing better machine utilization.
- *Sublevel caving is high production, low cost mechanized method.
- *Drill Machines can be operated remotely, reducing exposure of driller.
- *Workers only enter in stope after all blasting fume are exhausted by means of fan.
- *Large cross section of drift, raise, decline allows large ventilation air flow in mine.
- *Stope ventilation is good by auxiliary fan.

Advantages III

- *Man power has low exposure to unsafe, unprotected activities in stoping area.
- *Rate of production is very high.
- *Recovery of ore is high, up to 90%.
- *Very large machine can be deployed, giving efficient & economy to method.
- *Sublevel caving method is low cost method for underground metal mine.
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Disadvantages I

- *Sublevel caving is capital intensive (large development cost, especially when ore body dip is gentle less than 40 degree)
- *Sublevel caving is not selective, most of ore including waste has to be mined.
- *Sublevel caving is very inefficient at low dip (dilution goes up significantly).
- *Ore produced can be highly diluted by waste rock.
- *When extraction limit has been achieved. Remaining highly diluted ore represents ore loss.

Disadvantages II

- *Some ore is lost in passive zone located on level of extraction between active zone of gravity flow.
- *Mining generates progressive caving in over laying rock & results in subsidence & damage to surface.
- *Wider area on surface gets degenerated.
- *Development cost is 30% of total mining cost.

Disadvantages III

- *Some time ore is left in stope passive area.
- *Subsidence & caving occurs at wider area causing land degradation.
- *Ore dilution rate is high (upto 30%)
- *Development cost is high, up to 30% of mining cost.
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References I

- *SME Mining Engineering Hand Book, Hartman, 2nd Edition, Page 1789 to 1814.
- *Surface & Underground Excavation Method, Techniques and Equipment by Ratan Raj Tantiya, Page 460 to 466